

Novelty Comparison: Human-AI vs Human-Human vs AI-AI

Cross-Condition Novelty, Transience, and Resonance Analysis

PENPAL Project

2026-05-11

Contents

1	Overview	1
2	Load Data	2
3	Preprocessing	2
4	Primary: Within-Condition Slot Slope-Interaction Asymmetry	2
5	Bootstrap CIs and Effect Sizes on A_influence_story	3
6	Planned Contrast: HA vs (HH + AA)/2	4
7	Directional Stability of Signed Novelty-Resonance Slope Asymmetry	4
8	Length-Controlled Variant of A_influence	5
9	Supplemental: General Condition-Level Comparison	5
10	Supplemental: Pooled Novelty-to-Resonance Relationship	8
11	Supplemental: Role-Level View	10
12	Summary	10

1 Overview

This notebook compares three information-theoretic metrics across conditions:

- **Novelty / surprise**: how unexpected a contribution is given prior context
- **Transience**: how quickly its influence fades
- **Resonance**: novelty minus transience

The notebook now follows the new data structure:

- it loads novelty scores through the shared comparison helpers,
- it restricts the analysis to **complete exchanges** in the aligned turn window,
- it treats `author_1` and `author_2` as **slot 1 / slot 2** rather than fixed identities,
- it uses role labels only where they are structurally valid:
 - **Human / AI** in Human-AI
 - **Starter-side / Non-starter-side** in same-type conditions

- role-labeled plots are therefore **within-condition descriptive views**, not direct cross-condition comparisons

Primary cross-condition object. Following the paper’s narrative-influence model, the main inferential result is the **within-condition novelty-to-resonance slope interaction** between the two slots, fit as $\text{resonance} \sim \text{surprise} \times \text{slot} + (1 \mid \text{conversation_id})$ per condition. The cross-condition asymmetry magnitude is

$$A_{\text{influence}}(\text{condition}) = |\beta_{\text{surprise, slot=1}} - \beta_{\text{surprise, slot=2}}|,$$

extracted via **emtrends**. Per-story OLS slope differences provide a label-invariant robustness layer. Bootstrap CIs, Cliff’s δ effect sizes, the planned HA vs (HH+AA)/2 contrast, and a length-controlled re-fit are added below. The signed Human/AI decomposition lives in analysis/human-ai/novelty.Rmd and is not duplicated here. Starter-mechanism analysis is deferred.

Article-use note. The article-facing result is the magnitude of the novelty-to-resonance slope gap. Do not infer Human-AI specificity from within-condition significance alone; the direct cross-condition question requires comparing the slot-slope gap across conditions and checking the per-story absolute gap as a robustness layer.

2 Load Data

Table 1: Loaded novelty data by condition

condition	n_stories	n_turns
ai-ai	80	720
human-ai	96	763
human-human	36	324

3 Preprocessing

Slot-level rows: 3458

Metric rows: 10374

4 Primary: Within-Condition Slot Slope-Interaction Asymmetry

Table 2: Primary: $|\text{slope_1} - \text{slope_2}|$ from $\text{resonance} \sim \text{surprise} *$ slot per condition

condition	slope_slot1	slope_slot2	signed_diff	se	t	p	A_influence	sig
Human-AI	1.108	1.046	-0.061	0.034	-1.821	0.069	0.061	.
Human-Human	1.101	1.101	0.000	0.043	0.003	0.998	0.000	
AI-AI	1.155	1.194	0.039	0.026	1.510	0.131	0.039	

Table 3: Pooled model: $|\text{slope_slot1} - \text{slope_slot2}|$ (unsigned) by condition

condition	A_influence	se
Human-AI	0.051	0.033

condition	A_influence	se
Human-Human	0.001	0.044
AI-AI	0.043	0.026

Table 4: Robustness: per-story $|\text{slope}_1 - \text{slope}_2|$ (OLS per story, 4 points per slot)

condition	n_stories	median_A	mean_A
Human-AI	96	0.324	0.416
Human-Human	36	0.256	0.298
AI-AI	80	0.267	0.289

```
##
## Kruskal-Wallis on per-story A_influence:
##
## Kruskal-Wallis rank sum test
##
## data: A_influence_story by condition
## Kruskal-Wallis chi-squared = 4.2909, df = 2, p-value = 0.117
##
##
## Pairwise Wilcoxon on per-story A_influence (BH-adjusted):
##
## Pairwise comparisons using Wilcoxon rank sum test with continuity correction
##
## data: per_story_slope_diff$A_influence_story and per_story_slope_diff$condition
##
##           Human-AI Human-Human
## Human-Human 0.25      -
## AI-AI       0.16      0.87
##
## P value adjustment method: BH
```

5 Bootstrap CIs and Effect Sizes on A_influence_story

Table 5: Bootstrap 95% CI on per-condition mean of per-story A_influence (R = 10,000)

condition	n	mean	ci_lower	ci_upper
Human-AI	96	0.416	0.350	0.487
Human-Human	36	0.298	0.227	0.371
AI-AI	80	0.289	0.245	0.335

Table 6: Pairwise Cliff's delta on per-story A_influence

contrast	n_a	n_b	cliff_delta	magnitude
Human-AI vs Human-Human	96	36	0.157	small

contrast	n_a	n_b	cliff_delta	magnitude
Human-AI vs AI-AI	96	80	0.169	small
Human-Human vs AI-AI	36	80	0.019	negligible

6 Planned Contrast: HA vs (HH + AA)/2

Table 7: Planned contrast: per-story A_influence in HA vs the average of HH and AA

contrast	estimate	se	t	df	p	ci_lower	ci_upper
Human-AI vs (Human-Human + AI-AI)/2	0.123	0.042	2.943	149.016	0.004	0.04	0.205

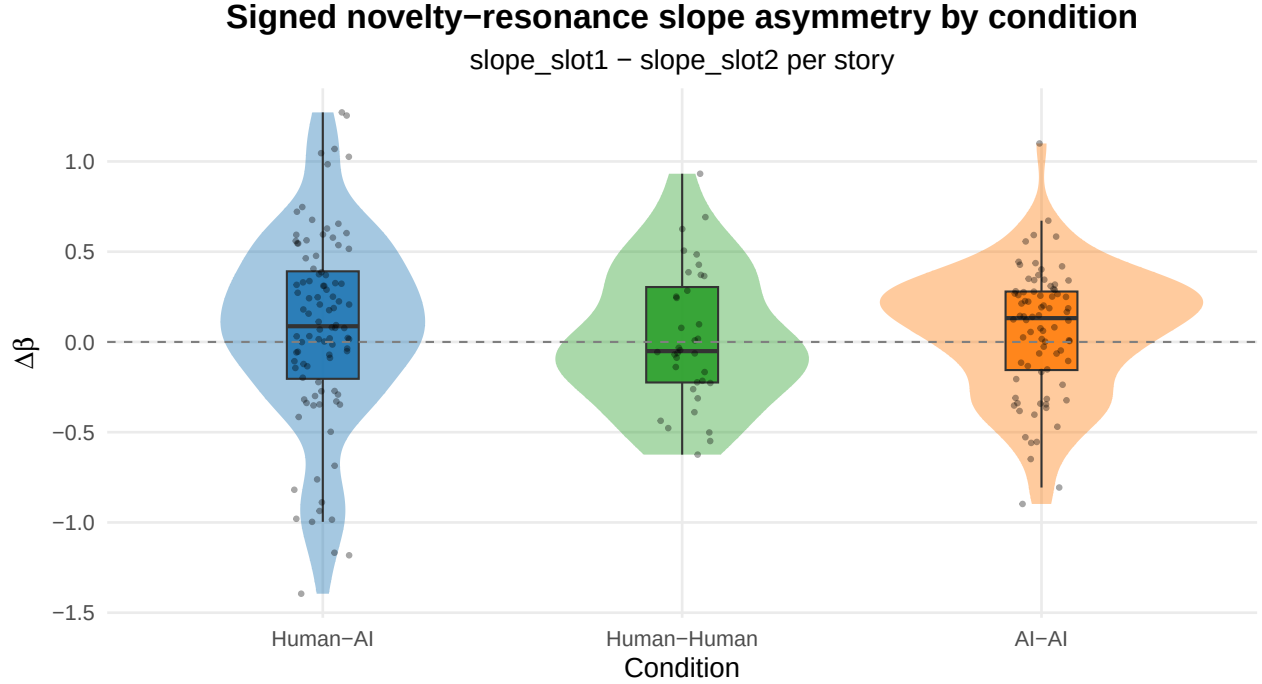
7 Directional Stability of Signed Novelty-Resonance Slope Asymmetry

Tests whether the *direction* of the slot-1 vs slot-2 novelty-to-resonance slope difference is consistent across stories within each condition. The signed quantity is $\text{signed}_i = \beta_i^{\text{slot1}} - \beta_i^{\text{slot2}}$. HA-specific identity-tied directionality predicts high stability in HA and low stability in HH/AA, even when unsigned **A_influence** is similar.

Table 8: Per-condition directional stability of signed novelty-resonance asymmetry

condition	n	mean	sd	stability	sign_consistency	one_sample_t_p
Human-AI	96	0.079	0.537	0.147	0.615	0.154
Human-Human	36	0.023	0.375	0.062	0.556	0.714
AI-AI	80	0.057	0.352	0.163	0.650	0.150

Brown-Forsythe variance-equality p = 0.0068



8 Length-Controlled Variant of $A_{\text{influence}}$

Re-fit the within-condition LMM with turn length as an additive covariate and recompute the unsigned slope-slot magnitude. Tests whether condition-level slot-asymmetry differences survive turn-length adjustment.

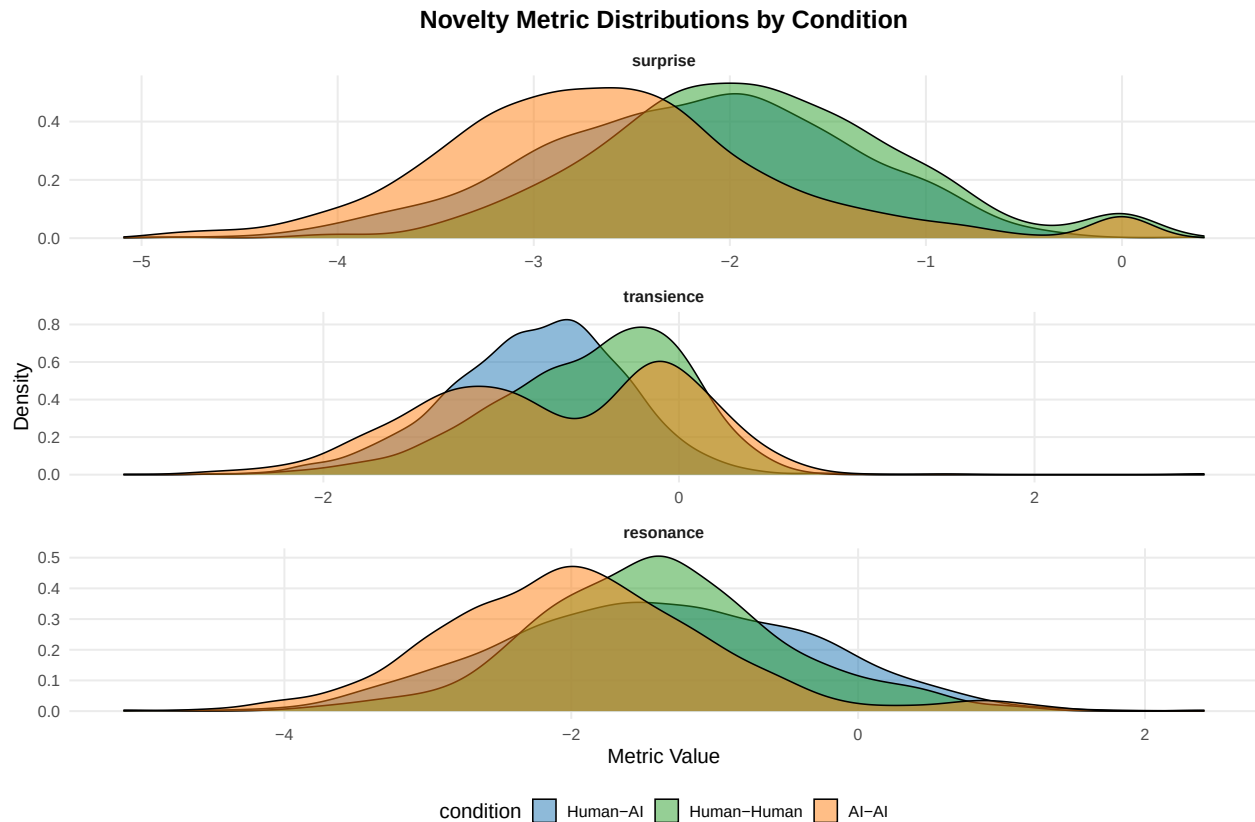
Table 9: Length-controlled $|\text{slope_slot1} - \text{slope_slot2}|$ (resonance ~ surprise * slot + nwords)

condition	$A_{\text{influence_lc}}$	se	p
Human-Human	0.001	0.044	0.984

9 Supplemental: General Condition-Level Comparison

Table 10: Overall novelty metrics by condition

condition	metric	n_stories	n_obs	mean_value	sd_value
Human-AI	surprise	96	1430	-2.193	0.796
Human-AI	transience	96	1430	-0.807	0.495
Human-AI	resonance	96	1430	-1.386	1.051
Human-Human	surprise	36	629	-1.887	0.771
Human-Human	transience	36	629	-0.494	0.547
Human-Human	resonance	36	629	-1.393	0.871
AI-AI	surprise	80	1399	-2.617	0.873
AI-AI	transience	80	1399	-0.682	0.718
AI-AI	resonance	80	1399	-1.935	0.975



```
##
## === surprise ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ condition + (1 | conversation_id)
## Data: filter(df_long, metric == metric_name)
##
## REML criterion at convergence: 8113.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.6538 -0.6292 -0.0281  0.6050  3.6481
##
## Random effects:
## Groups      Name                Variance Std.Dev.
## conversation_id (Intercept) 0.1214   0.3485
## Residual                0.5559   0.7456
## Number of obs: 3458, groups: conversation_id, 212
##
## Fixed effects:
##              Estimate Std. Error    df t value Pr(>|t|)
## (Intercept)   -2.19383    0.04067 220.70480 -53.942 < 2e-16 ***
## conditionHuman-Human    0.30044    0.07689 209.71158   3.908 0.000126 ***
## conditionAI-AI       -0.43135    0.05975 212.39417  -7.220 9.13e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
```

```

##          (Intr) cndH-H
## cndtnHmn-Hm -0.529
## condtnAI-AI -0.681  0.360

## contrast          estimate      SE  df z.ratio p.value
## (Human-AI) - (Human-Human) -0.300 0.0769 Inf  -3.908  0.0003
## (Human-AI) - (AI-AI)      0.431 0.0597 Inf   7.220  <.0001
## (Human-Human) - (AI-AI)    0.732 0.0786 Inf   9.314  <.0001
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: bonferroni method for 3 tests
##
## === transience ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ condition + (1 | conversation_id)
## Data: filter(df_long, metric == metric_name)
##
## REML criterion at convergence: 4802
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.9949 -0.5657  0.0654  0.6061  8.3614
##
## Random effects:
## Groups          Name          Variance Std.Dev.
## conversation_id (Intercept) 0.1585   0.3981
## Residual                0.1992   0.4463
## Number of obs: 3458, groups: conversation_id, 212
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   -0.80622    0.04232 213.55480 -19.052 < 2e-16 ***
## conditionHuman-Human  0.30328    0.08069 209.99226  3.759 0.000221 ***
## conditionAI-AI      0.10939    0.06257 210.86688  1.748 0.081859 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##          (Intr) cndH-H
## cndtnHmn-Hm -0.524
## condtnAI-AI -0.676  0.355

## contrast          estimate      SE  df z.ratio p.value
## (Human-AI) - (Human-Human) -0.303 0.0807 Inf  -3.759  0.0005
## (Human-AI) - (AI-AI)      -0.109 0.0626 Inf  -1.748  0.2412
## (Human-Human) - (AI-AI)    0.194 0.0827 Inf   2.344  0.0573
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: bonferroni method for 3 tests
##
## === resonance ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ condition + (1 | conversation_id)

```

```
## Data: filter(df_long, metric == metric_name)
##
## REML criterion at convergence: 9747.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.8116 -0.6717 -0.0387  0.6094  4.5646
##
## Random effects:
##   Groups             Name             Variance Std.Dev.
## conversation_id (Intercept) 0.0152    0.1233
## Residual                  0.9641    0.9819
## Number of obs: 3458, groups: conversation_id, 212
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    -1.386408   0.028857 248.399233  -48.04  <2e-16 ***
## conditionHuman-Human -0.006353   0.052802 207.377682   -0.12   0.904
## conditionAI-AI    -0.547021   0.041376 217.161808  -13.22  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) cndH-H
## cndtnHmn-Hm -0.547
## condtnAI-AI -0.697  0.381
##
## contrast            estimate      SE  df z.ratio p.value
## (Human-AI) - (Human-Human) 0.00635 0.0528 Inf   0.120  1.0000
## (Human-AI) - (AI-AI)       0.54702 0.0414 Inf  13.221  <.0001
## (Human-Human) - (AI-AI)    0.54067 0.0532 Inf  10.155  <.0001
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: bonferroni method for 3 tests
```

10 Supplemental: Pooled Novelty-to-Resonance Relationship

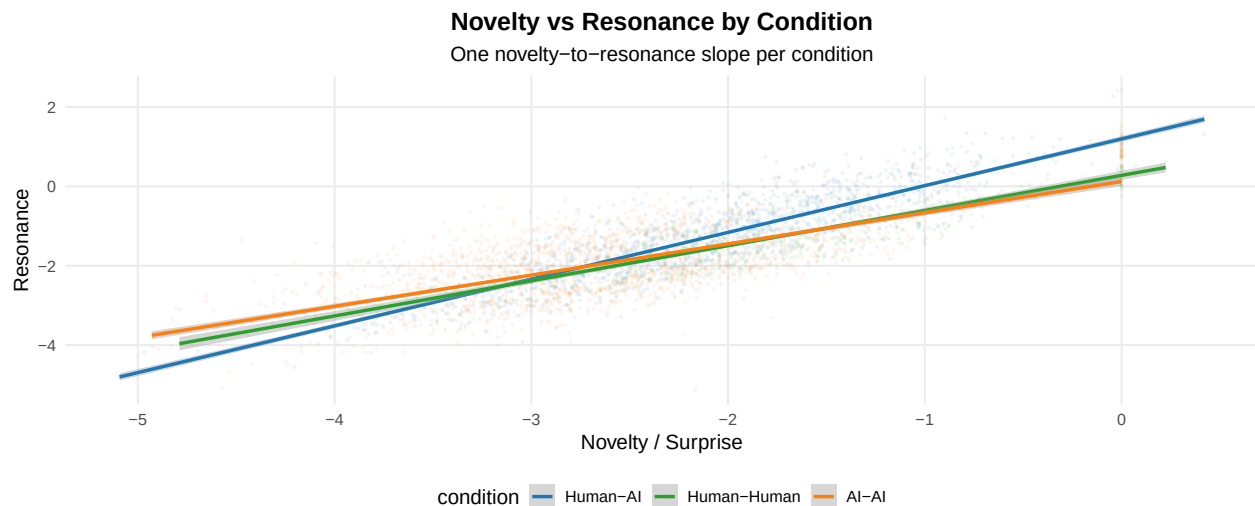
```
## Novelty-to-resonance model:
##
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: resonance ~ surprise * condition + (1 | conversation_id)
## Data: df_slots
##
## REML criterion at convergence: 4499.6
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -8.8382 -0.6157 -0.0167  0.5726  4.5150
##
## Random effects:
##   Groups             Name             Variance Std.Dev.
## conversation_id (Intercept) 0.1970    0.4439
## Residual                  0.1783    0.4223
```



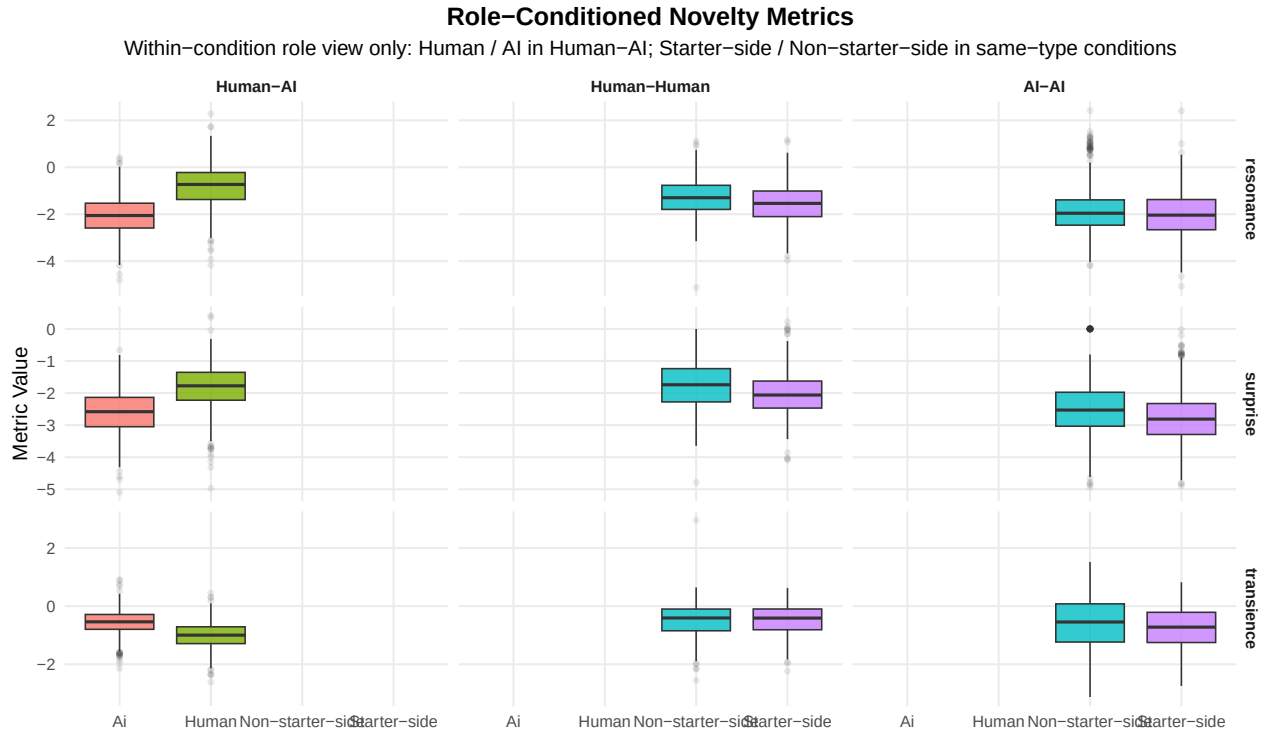
```

## Number of obs: 3458, groups: conversation_id, 212
##
## Fixed effects:
##
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      1.33323    0.05701  448.89437   23.387  < 2e-16
## surprise          1.24020    0.01493 3287.90417   83.091  < 2e-16
## conditionHuman-Human -0.64504    0.10635  404.12971  -6.065 3.03e-09
## conditionAI-AI      -0.29404    0.08644  483.32980  -3.402 0.000725
## surprise:conditionHuman-Human -0.14252    0.02940 3329.19779  -4.848 1.30e-06
## surprise:conditionAI-AI -0.10997    0.02143 3327.26141  -5.132 3.03e-07
##
## (Intercept)      ***
## surprise          ***
## conditionHuman-Human ***
## conditionAI-AI      ***
## surprise:conditionHuman-Human ***
## surprise:conditionAI-AI ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) surprs cndH-H cAI-AI sr:H-H
## surprise      0.574
## cndtnHmn-Hm -0.536 -0.308
## condtnAI-AI -0.659 -0.379  0.354
## srprs:cnH-H -0.292 -0.508  0.545  0.192
## srprs:AI-AI -0.400 -0.697  0.214  0.599  0.354
##
##
## Condition-specific novelty-to-resonance slopes:
## condition surprise.trend SE df asymp.LCL asymp.UCL
## Human-AI      1.24 0.0149 Inf      1.21      1.27
## Human-Human    1.10 0.0253 Inf      1.05      1.15
## AI-AI          1.13 0.0154 Inf      1.10      1.16
##
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95

```



11 Supplemental: Role-Level View



12 Summary

Primary cross-condition result: within each condition, $A_{\text{influence}} = |\beta_{\text{surprise, slot=1}} - \beta_{\text{surprise, slot=2}}|$ from `resonance ~ surprise * slot + (1|conversation_id)`. The pooled LMM contributes only the unsigned magnitude column. Per-story OLS slope differences provide the label-invariant robustness test.

Robustness layers added:

- bootstrap 95% CIs on per-condition mean of per-story **A_influence**
- pairwise Cliff's δ effect sizes
- planned single-df contrast HA vs (HH+AA)/2
- length-controlled re-fit (+ **nwords**) of the unsigned magnitude

Supplemental views retained:

- condition-level marginal distributions and pooled **condition** LMMs
- pooled novelty-to-resonance model with **surprise * condition** interaction
- within-condition role box-plots (descriptive only; HA Human/AI inferential view lives in `analysis/human-ai/novelty.Rmd`)

Deferred to a follow-up pass:

- starter-mechanism analysis (signed, starter-normalized asymmetry across all conditions)

Analysis completed: 2026-05-11 16:28:58.932345